

Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.

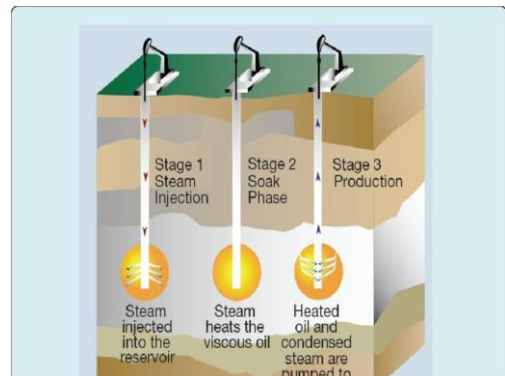
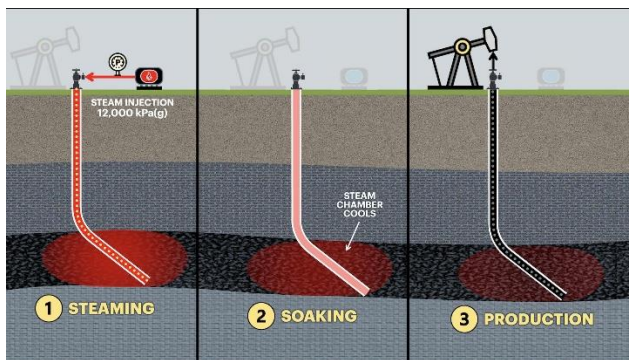
Cyclic Steam Stimulation

💡 What is Cyclic Steam Stimulation?

Cyclic Steam Stimulation, often called the “Huff-and-Puff” method, is a thermal oil-recovery technique.

The basic idea is:

Inject steam → Wait → Produce oil → Repeat



1. ☞ HUFF — Steam Injection

First, **high-temperature steam is injected into the oil well.**

The steam travels from the well into the surrounding oil-bearing reservoir.

Why do we do this?

Heavy oil is very **viscous**, meaning it is thick and doesn't flow easily—similar to how **honey flows much more slowly than water**.

The hot steam transfers heat to the heavy oil, causing its **viscosity to decrease**.

So:

Steam → Heat → Oil temperature increases → Oil becomes less viscous → Oil flows more easily

2. ⌚ SOAK — Let the Heat Spread

After injecting enough steam, the well is temporarily **shut in**.

This is called the **soak period**.

During this time, we don't produce oil. Instead, we allow the heat from the steam to **spread through the reservoir**.

You can imagine putting a spoonful of hot water into thick honey and waiting for the heat to spread. The honey near the heat source becomes thinner and easier to move.

3. **PUFF — Produce the Oil**

Now the well is opened for production.

Because the oil has been heated:

- Its **viscosity decreases**
- It can flow more easily toward the well
- Reservoir pressure and gravity help move the oil
- The heated oil is produced to the surface

This production stage is called the **Puff**.

Why is it called "Cyclic"?

After production decreases, the same process can be repeated:

Huff → Soak → Puff → Huff → Soak → Puff → ...

Each complete sequence is called a **cycle**.

The easiest way to remember it

Think of **CSS** like warming up thick honey:

Heat it → wait for the heat to spread → let it flow → repeat.

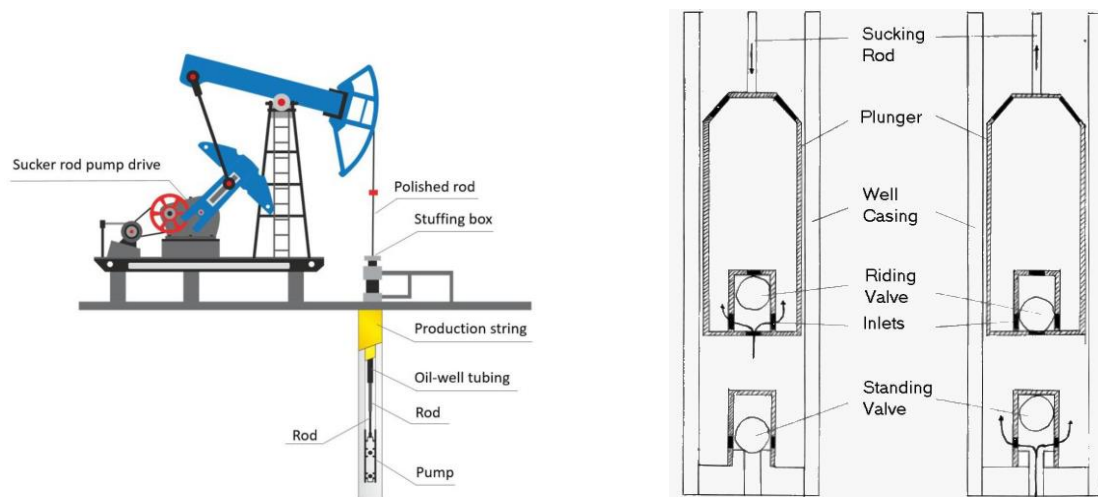
The key principle is **thermal viscosity reduction**. We aren't primarily making the oil disappear or chemically changing it; we're **heating it so that it becomes much less resistant to flow**.

In one sentence:

Cyclic Steam Stimulation (CSS) is a thermal enhanced oil recovery technique in which steam is periodically injected into an oil reservoir (Huff), the well is temporarily shut in to allow heat to penetrate the formation (Soak), and the heated, lower-viscosity oil is subsequently produced from the well (Puff).

Sucker Rod Pump

What is a Sucker Rod Pump (SRP)?



A **Sucker Rod Pump (SRP)** is a mechanical system used to **bring crude oil from deep underground to the surface** when the natural pressure of the oil reservoir is no longer strong enough to push the oil up by itself.

You can think of it as a **long mechanical plunger system inside an oil well**.

First, imagine a simple water well

Suppose you have a deep well containing water. You put a hand pump on top and repeatedly move the handle up and down. Inside the well, that movement operates a mechanism that pulls water upward.

An **SRP works on a similar principle**, except:

- The well can be **thousands of metres deep**.
- The fluid is usually **crude oil**, which can be much more viscous than water.
- Instead of a person operating the handle, a **motor** operates the pump.
- A long string of **sucker rods** transfers the mechanical movement from the surface down into the well.

1. Why do we need an SRP?

When an oil well is first drilled, the underground reservoir may have enough pressure to push oil naturally toward the surface.

This is called **natural flow**.

For example:

Reservoir → Oil rises through well → Surface

But as oil is produced, the reservoir pressure can decrease.

Eventually:

Reservoir pressure ↓ → Oil cannot rise to surface efficiently

So we need an **artificial lift system**.

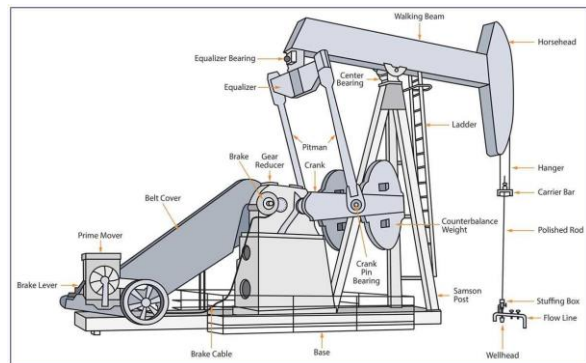
One of the most common artificial-lift methods is the **Sucker Rod Pump (SRP)**.

Its basic job is:

Take mechanical energy from the surface and use it to pump oil from deep underground to the surface.

2. What does the SRP actually look like?

You have probably seen something like this near oil fields:



That large machine is commonly called a **pumpjack**.

You may also hear names such as:

- Pumpjack
- Beam pump
- Walking beam pump
- Nodding donkey

The important thing to understand is:

The pumpjack that you see on the surface is NOT the actual pump.

It is the **surface mechanism that drives the downhole pump**.

3. The SRP has two main parts

It's easiest to understand SRP by dividing it into:

A. Surface equipment

Located above the ground.

B. Downhole equipment

Located deep inside the well.

The two are connected by a long string of **sucker rods**.

4. What are sucker rods?

This is probably the most important thing to understand from the name **Sucker Rod Pump**.

A **sucker rod** is a long, strong metal rod that connects the surface pumping unit to the pump located deep underground.

Imagine a **very long metal stick** going down into the oil well.

The surface machine moves this rod:

UP ↓ DOWN ↓ UP ↓ DOWN

That movement is transferred all the way to the pump at the bottom.

So:

Sucker rods transmit the reciprocating mechanical motion from the surface to the downhole pump.

"Reciprocating" simply means:

back-and-forth motion.

5. How does the pump actually lift oil?

This is the really important part.

Inside the downhole pump, there is a **plunger** and valves.

There are two important valves:

1. Standing valve

Usually remains at the bottom of the pump.

2. Travelling valve

Moves with the plunger.

These valves make sure that oil moves **upward** instead of simply moving back and forth.

6. Let's understand one complete pumping cycle

This is the easiest way to understand SRP.

The pump goes through two main strokes:

Downstroke

The plunger moves **down**.

Upstroke

The plunger moves **up**.

Let's see what happens.

▼ Step 1: Downstroke

The sucker rod and plunger move downward.

During this movement, the **travelling valve opens**, while the standing valve is closed/open depending on the pressure relationship, allowing fluid to move through the plunger into the space above it.

The key idea:

The pump prepares the fluid for the next lifting stroke.

▲ Step 2: Upstroke

Now the sucker rod pulls the plunger **upward**.

The travelling valve closes.

The standing valve opens, allowing new reservoir fluid to enter the pump.

At the same time, the fluid above the plunger is pushed upward.

So each upstroke moves a quantity of oil upward.

The process repeats:

Downstroke → Upstroke → Downstroke → Upstroke → ...

Thousands of times.

Eventually the oil reaches the surface.

8. But how does the surface motor create the up-and-down movement?

This is another important part.

An electric motor rotates continuously.

But the sucker rod needs to move **up and down**.

So the surface equipment converts:

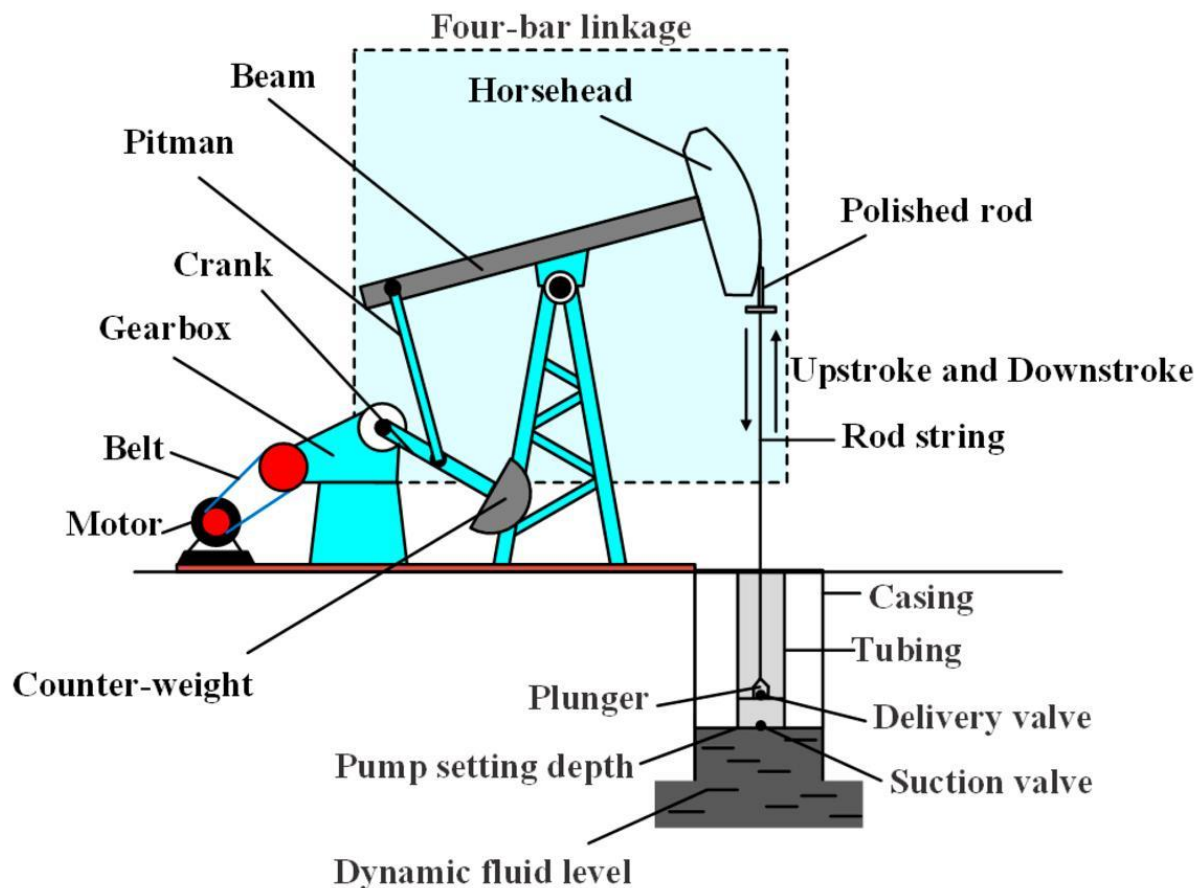
Rotary motion → Reciprocating motion

This is the heart of the SRP system.

9. What is the walking beam?

Look at a pumpjack.

The large horizontal beam that moves up and down is called the **walking beam**.



It acts somewhat like a **seesaw**.

One side is connected to the crank mechanism.

The other side moves the sucker rods.

So when the motor rotates the crank:

Crank rotates → walking beam rocks → sucker rod moves up and down

10. What is the polished rod?

At the very top of the sucker-rod string is a component called the **polished rod**.

It passes through a sealing arrangement at the wellhead called the **stuffing box**.

The polished rod connects the surface pumping unit to the sucker rods below.

You can think of it as:

Walking beam → Polished rod → Sucker rods → Downhole pump

11. Why is it called a "Sucker Rod Pump"?

Because the **sucker rod string** operates the downhole pump.

So the name basically tells you:

Sucker Rod + Pump = Sucker Rod Pump

The rods don't themselves "suck" the oil.

Instead, they **transmit mechanical movement to the pump**, and the pump lifts the fluid.

12. A very simple analogy

Imagine a **bicycle hand pump** or a **syringe**.

If you repeatedly move a plunger:

Pull → Push → Pull → Push

and use valves correctly, you can move fluid in one direction.

An SRP is essentially a **much larger and much deeper mechanical pumping system**, with:

- Motor
- Crank mechanism
- Beam
- Rod string
- Plunger
- Valves

working together.

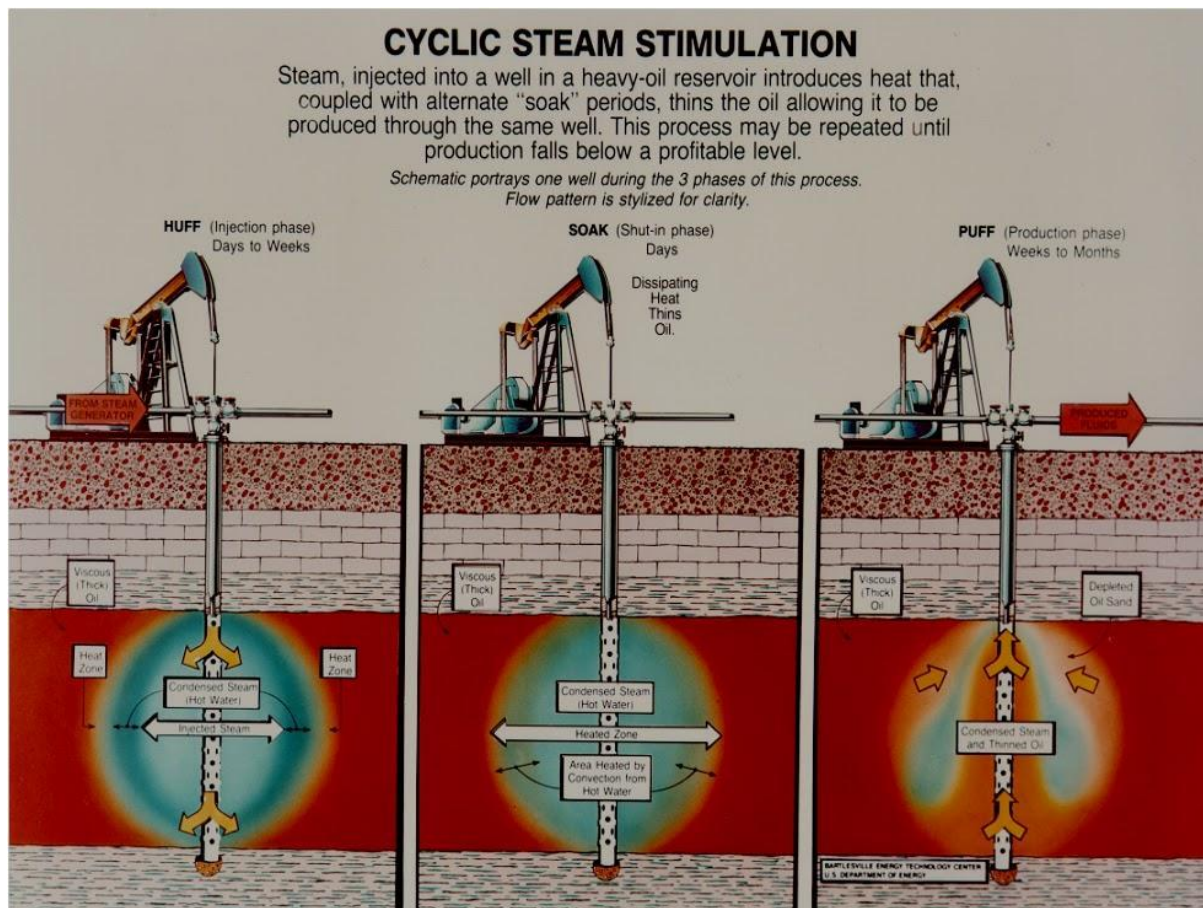
13. The entire process in one picture

The motor provides energy, the surface mechanism converts rotary motion into up-down motion, the sucker rods transmit that motion, and the downhole pump uses it to lift oil.

The Core Conflict

1. First, understand the overall process

Imagine a heavy-oil well like this:



There are basically two important operations:

CSS → heats the underground oil

Steam is injected into the reservoir. The heat reduces the oil's viscosity, making the heavy crude flow more easily.

SRP → brings the oil to the surface

A sucker rod pump is a mechanical pump installed down the well. A motor at the surface moves the sucker rods up and down, which operates the pump and lifts oil to the surface.

So, very simply:

Steam → makes oil easier to flow → Pump → brings oil to surface

2. What does "Decoupled Management" mean?

Your first point says:

CSS cycle timing and SRP operational settings are tuned in silos.

Decoupled simply means **they are being controlled independently**.

For example, suppose an operator decides:

- Inject steam for 5 days
- Wait for 2 days

- Start pumping
- Run SRP at 6 SPM
- Use a certain stroke length

The problem is that the **CSS system isn't necessarily considering what the SRP is doing**, and the **SRP isn't necessarily considering the current thermal condition of the reservoir**.

Think of it like two people driving the same car but **not talking to each other**.

CSS controller thinks:

"The reservoir needs more steam, so I'll inject more steam."

Pump controller thinks:

"We need more oil production, so I'll increase SPM."

But these decisions are connected.

If the reservoir is very hot, the oil may be relatively thin → the pump can operate aggressively.

If the reservoir has cooled down, the oil becomes extremely thick → the pump may struggle if operated at the same speed.

Therefore:

CSS conditions should influence SRP settings, and SRP performance should influence CSS decisions.

That's the fundamental idea behind the conflict.

3. Now the important part: The Cooling Phase Dilemma

This is probably the most important concept in your problem.

After steam injection stops, the reservoir doesn't remain hot forever.

The sequence is roughly:

Steam injection → Hot reservoir → Steam stops → Reservoir cools → Oil becomes more viscous

Remember:

Temperature ↑ → Oil viscosity ↓

and

Temperature ↓ → Oil viscosity ↑

Viscosity is basically a measure of **how difficult it is for a fluid to flow**.

Think about:

Water □

It flows very easily.

Compared with:

Honey 🍯

Honey flows much more slowly because it has much higher viscosity.

Heavy crude oil can behave more like very thick syrup/honey than water.

4. What happens when the reservoir cools?

During the hot phase:

Hot oil → low viscosity → easier movement

But after steam injection:

Cooling → viscosity increases → oil becomes harder to move

Now the SRP has a problem.

The pump is trying to move the sucker rod string up and down while dealing with increasingly viscous oil.

And this creates something called **rod floating**.

5. What is rod floating?

Don't imagine the rod literally floating in the oil.

It means that the **sucker rod string does not follow the intended motion smoothly**.

Normally, the surface motor tells the sucker rod:

DOWN ↓

and the rod should move down smoothly.

But when the crude becomes extremely viscous, the pump and fluid resistance can interfere with that motion.

The rod may temporarily **lag behind the intended movement**, creating slack in the rod string.

Think about a long rope.

If you move one end quickly, the other end doesn't necessarily respond instantly.

A sucker rod string is somewhat similar, except it is a long mechanical system operating deep underground.

6. What is mechanical slack?

Suppose the controller commands:

Rod → DOWN ↓

But because of the high resistance from viscous crude, part of the rod string doesn't move as expected.

You can get something like:

Commanded movement:

↓↓↓↓

Actual movement:

↓ ... ↓ ... ↓

The rod string can temporarily lose proper tension.

This is called **slack**.

Then, when the rod string suddenly becomes loaded again, the stored mechanical energy can produce a large impact.

7. Why is high impact loading bad?

Imagine dropping a hammer.

If you slowly place a hammer on something:

small force

But if you drop it:

large impact force

The same basic idea applies here.

If the sucker rod string develops slack and then suddenly becomes loaded, you can get a **large impact load**.

Repeated impacts are very damaging to mechanical components.

They can contribute to:

- sucker rod fatigue
- rod/tubing contact
- pump problems
- mechanical wear
- premature failure

This is why the cooling phase is dangerous for the SRP.

8. What does "pump unsetting" mean?

The pump needs to operate with the correct mechanical loading and movement.

If the rod movement becomes irregular because of excessive fluid resistance and dynamic loading, the pump may not operate under its intended conditions.

This can lead to **pump unsetting**, meaning the pump's intended operating/loading condition is disturbed.

In simple terms:

The pump is no longer working in the stable mechanical condition it was designed to operate in.

That can reduce production and increase the chance of failure.

9. What is rod/tubing fatigue?

Inside the well, you have:

Tubing

and inside/around it:

Sucker rods

The sucker rods repeatedly move up and down.

So the system experiences millions of repeated loading cycles.

If the loads are smooth:

Normal loading → normal fatigue life

But if you introduce repeated high-impact loads:

Impact → impact → impact → impact

the components can fatigue much faster.

Eventually, a rod can develop cracks or fail.

So your chain becomes:

Reservoir cools

↓

Oil viscosity increases

↓

Higher resistance to pumping

↓

Rod floating / slack

↓

Impact loading

↓

Pump and rod/tubing problems

↓

Higher maintenance + lower production

10. Now let's understand the Energy Inefficiency problem

There are actually **two competing problems**.

Problem A — Too much steam

Suppose the reservoir already has enough heat.

But you continue injecting large amounts of steam.

You're spending energy to generate and inject that steam without getting enough additional oil production.

This increases:

Steam-Oil Ratio (SOR)

Very simply:

$$\text{SOR} = \text{Steam used} / \text{Oil produced}$$

So if you use a lot of steam but produce relatively little oil:

SOR ↑

A high SOR generally indicates poorer steam utilization.

11. Problem B — Pumping too aggressively

Now imagine the reservoir has cooled and the oil has become extremely viscous.

Instead of reducing the pump speed, suppose you continue operating the SRP aggressively:

High SPM

Large stroke

High VFD speed

The pump is demanding a lot from the mechanical system while the fluid is very difficult to move.

This can lead to:

- higher motor power consumption
- increased mechanical loads
- rod/tubing stress
- higher impact loading
- inefficient production

So you're using more electricity without necessarily getting proportionally more oil.

12. What are SPM, stroke length and VFD speed?

These are important because you'll probably see them repeatedly in your project.

SPM — Strokes Per Minute

How many complete up-and-down cycles the pump performs per minute.

For example:

6 SPM = 6 pumping cycles per minute

Higher SPM generally means faster pumping.

Stroke Length

How far the sucker rod moves during each stroke.

For example:

Stroke = 2 m

means the rod travels approximately 2 m during one pumping stroke.

Increasing stroke length can increase the amount of fluid displaced per cycle, depending on the pump and operating conditions.

VFD Speed

VFD = Variable Frequency Drive

A VFD controls the speed of an electric motor by controlling the electrical frequency supplied to it.

So:

VFD setting ↑ → Motor speed ↑ → Pumping speed can ↑

Therefore, VFD speed is another way of controlling how aggressively the SRP operates.

13. So what is the REAL conflict?

Now we can put everything together.

You have **two systems**:

System 1 — CSS

Controls the **thermal condition of the reservoir**.

Steam injection:

Steam → Heat → Lower viscosity → Easier oil flow

System 2 — SRP

Controls the **mechanical production of oil**.

Pump:

Motor → Rod movement → Pump → Oil production

The problem is that these two systems are **strongly coupled physically**, but they are being **managed separately**.

The problem can be visualized like this:

CSS

Steam injection



Reservoir temperature



Oil viscosity



Oil flowability

SRP

VFD speed



SPM



Stroke length



Pump movement



Oil production

But the two chains are connected:

Reservoir temperature ↓

→ **Viscosity ↑**

→ **Pump resistance ↑**

→ **Rod dynamics change**

→ **Impact loading ↑**

→ **Mechanical problems ↑**

At the same time:

Too much steam

→ **Energy consumption ↑**

→ **SOR ↑**

And:

Too aggressive pumping

→ **Electricity consumption ↑**

→ **Mechanical stress** ↑

14. What our project is essentially trying to solve

The ideal system would **coordinate CSS and SRP together**.

Instead of saying:

"Let's inject steam according to the old schedule."

and separately:

"Let's run the pump at 6 SPM."

the system should ask:

"What is the current reservoir temperature, oil viscosity, production rate and mechanical condition, and therefore what should the steam injection and SRP settings be right now?"

That is the key concept.

Instead of fixed/rule-of-thumb operation:

CSS schedule + SRP settings

↓

Independent decisions

you want:

Coordinated optimization:

Reservoir temperature + pressure + viscosity + production + rod load + pump condition

↓

Intelligent control

↓

Steam rate + CSS timing + SPM + stroke length + VFD speed

↓

Higher oil production + lower SOR + lower energy consumption + reduced mechanical damage